ADDIS ABABA INSTITUTE OF TECHNOLOGY
CIVIL AND ENVIRONMENTAL ENGINEERING/SCHOOL

Engineering Surveying Laboratory

Manual for Engineering Surveying I



Prepared by

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Preface

For any Civil Engineering project like building a house, constructing a dam, laying out a road, setting up of an industry etc., the first requirement is to have a plan/map of the area. To have a plan/map of any area, it is required to collect information and data about the terrain as well as the objects present in the area by taking necessary measurements using different types of instruments. Data thus collected are being subsequently used to prepare the plan/map of the area. The domain of engineering which involves the collection of field data and subsequently preparation of plan/map of the area is termed as 'Surveying'. Surveying is defined as the science of making measurements of the earth specifically the surface of the earth. This is being carried out by finding the spatial location (relative/absolute) of points on or near the surface of the earth. Different methods and instruments are being used to facilitate the work of surveying.

Surveying is primarily utilized to fix the national and state boundaries, chart coastlines, navigable streams and lakes, establishing control points, execute hydrographic and oceanographic charting and mapping, prepare topographic map of land surface of the earth, prepare plan or map of the area surveyed, collect field data, analyze and to calculate the field parameters for setting out operation of actual engineering works. Moreover, during execution, project of any magnitude is constructed along the lines and points established by surveying.

Thus, surveying is a basic requirement for all Civil Engineering projects. Based upon the consideration of the shape of the earth, surveying is broadly classified as geodetic surveying and plane surveying. Most of the civil engineering works, concern only with a small portion of the earth which seems to be a plane surface. Based on the purpose for which surveying is being conducted, surveying has been classified into: control surveying, land surveying, topographic surveys, engineering surveys, route surveys, construction surveys, astronomic surveys and mine surveys. Engineering survey is of prime importance consisting of three broad steps: reconnaissance survey, preliminary survey, location survey; which helps in collecting requisite data for planning, design and execution of engineering projects such as construction of highways, bridges, tunnels, dams etc. The primary aims of field surveying are to measure horizontal distance, vertical elevation and relative direction of lines by measuring horizontal angles. In the present day scenario, it has become mandatory for every civil engineer to have the basic awareness of surveying. Identifying this need, the university has introduced this practice based course for core branch of Civil Engineering at the second year level for undergraduate students.

The main objective of this study is to help students in gaining the practical experience by exposing them to various techniques of field surveying. The students will have an understanding of the concepts involved in the measurement of distance, angle/direction, and height at different instruments and techniques. At this juncture, the present course on Engineering Surveying Lab - I plays a vital role for enhancing the knowledge of an aspiring civil engineer. This lab course comprises of eight exercises which are intended to make the students to understand and gain familiarity with latest surveying techniques. The study consists of chain surveying, leveling, distance measurement using theodolite and EDM, angle measurement using Theodolite/Total Station,

detail measurement using Total Station, Trigonometric surveying and control establishment using traversing, and intersection and resections. .

At the end of this course, a student should be able to

- appreciate the role of the surveying in the civil engineering industry:
- understand how to perform basic survey measurements
- manage to establish control points using traversing and other methods
- manage, organize and execute a given task to meet specifications within a strict deadline;
- adapt to work in groups.

General

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General Information

Safety Guidelines

- Don't expose yourself to laser beam of total stations
- Never use a total station for a solar observation unless an approved solar filter is used. This will destroy an element in the EDM, plus damaging the eye of the observer.
- Transport and store instruments in positions that are consistent with the carrying case design.
- The person or group handling the instrument is responsible for any loss or damage.
- Keep the instruments in the case or mounted on the tripod. Never put them on the ground.
- Handle the instruments with care; never apply force when changing batteries, setting in memory cards etc. Always follow the instructions and figures to do the above named procedures.
- If the instruments are wet, open the case when storing them.

Report Requirements

- Use the word processor function for spelling check, cross references for literature, tables, figures etc in Microsoft Word to write the report.
- Structure the content of your report.
- **Remember that the exact structuring is determined by the nature of your work.**

1 **Cover Page:** Title of the field practice, Name, ID and Date.

2 **Abstract.** This should always be in English and it should be approximately 200 words. It should contain a brief description of the most important parts/problem formulations and

results in your report. It must be readable as a stand-alone part, so you do not normally have references in an abstract.

3 **Table of Contents.**

4 **Introduction.** State what your study is about, what you have investigated, the goals of the investigation, and why it is interesting.

5 **Background material.** A critical review of what has been done before in the area and how it relates to your work.

6 **Problem formulation**. A more careful formulation of the problem(s) investigated.

7 **Choice of methodology**. What methods did you choose to solve your problem and why? Are there other alternatives?

8 **Results.** Present and discuss critically the results you found. What conclusions can be made? Motivate what you write!

9 **Conclusions**. Sum up your most important results. Is there something general to say about them? What impact might they have? How can you move on?

10 **References**: A list of the sources you used.

About tables and figures

- Each table should be numbered and have a text that describes what it contains.
- Text associated to a figure must be under the figure while text concerning a table should be placed above the table.
- The first reference to a figure/table in the text should always come before the figure/table itself in the document.
- You must reference each figure and table in your text and describe what is said by the figure/table in words in your text and add a reference such as for instance: "*Figure 4 illustrates the mean queue length as a function of XXX. From these results and Table 1 it is evident that ...*"
- It must be possible to interpret each figure together with its figure text without forcing the reader to read all text that describes it. It English you say that the figures/tables should be "self-contained". (In practice this can be hard to guarantee. After all, words can explain more than a mere figure. If not, humans might not have seen a need to invent language...)
- A figure/table should always have a purpose. Never insert a figure/table in the report just to fill the space!

About reference lists

- Start the reference list on a separate page with the heading *References*.
- Order the references alphabetically based on the last name of the first author of each reference. If the same author has written several references they are sorted chronologically.
- Use an *italic* style for names of books and journals.
- For each reference the following must be included:
    - The full first and last names of the author (initials can be used for the first name). If there is no named author, for instance for a manual of a system, you can of course not name the author.
    - Publication year
    - Title of the work
    - Where is is published
    - The name of the editor/publisher (not for journals)
- Example of reference for books, articles and manuals are seen below

Book (monograph)

Wolf, P. R. and Ghilani, C. D., (2012), Elementary Surveying: An introduction to Geomatics, 13th ed, Prentice Hall.

Article

Jones, G. C., (2002), 'New solutions for the geodetic coordinate transformation', Journal of Geodesy, Vol. 76, pp. 437-446.

Manual, dictionary etc.

Guide to Parallel Programming on the Sequent Computer Systems. 1986, ISBN 0-13-370446-7, New York: Prentice-Hall

About quotations

- If you quote a source the quotations should be repeated word-by-word and enclosed with " ".
- If you leave out part of a quotation, if is too long or irrelevant in the context, you must mark this with three dots: ...

- If you add comments or explanations, you should put them in brackets [].
- Normally you do not quote more than 100 words in running text. E.g.: *"This method [to synchronize a parallel simulation] is primarily suited for large scale systems. ... and has been found to work well for the simulation of mini-busses" (Unknown 1975:123)*. In the example "*to synchronize a parallel simulation*" is added as a comment/explanation and the irrelevant parts are skipped and marked with ...

Exercise 1

1 Chain Surveying

1.1 Objective of the Exercise

The objective of this exercise is to make you familiar with taping, ranging, compass and right angle measurement techniques. After completing the exercise you should be able to apply these techniques in real life situations and carry out detailed surveying on smaller areas.

1.2 Theory

In this method, steel tapes or wires are used to measure distance very accurately. Nowadays, EDM is being used exclusively for accurate measurements but the steel tape still is of value for measuring limited lengths for setting out purposes.

Tape measurements require certain corrections to be applied to the measured distance depending upon the conditions under which the measurements have been made. These corrections are summarized in Table 1-1.

Table 1-1 Tape Corrections

Correction	Sign	Formula
Absolute length (c_a)	±	$\frac{c}{l}L$
Temperature (c_t)	±	$\alpha(t_m - t_0)L$
Pull (c_p)	±	$\frac{(P-P_0)}{AE}L$
Sag (c_g)	–	$\frac{1}{24}\left(\frac{W}{P}\right)^2 L$
Slope (c_s)	–	$(1-\cos\theta)L$ *(exact)*
Alignment (c_m)	–	$\frac{h^2}{2L}$ *(approximate)*
Mean sea level (c_{msl})	–	$\frac{d^2}{2L}$ *(approximate)*
		$\frac{HL}{R}$ *(approximate)*

1.3 Instrument Used

a) Chain

The chain is composed of 100 or 150 pieces of galvanized mild steel wire, 4mm in diameter called links .The ends of each link are bent into a loop and connected together by means of three oval rings. The ends of the chain are provided with handles for dragging the chain on the ground, each wire with a swivel joint so that the chain can be turned without twisting. The length of the chain is measured from the outside of one handle to the outside of another handle.

Following are the various types of chain in common use:
1) Metric chains
2) Gunter` s chain or surveyors chain
3) Engineers chain
4) Revenue chain
5) Steel band or Band chain

b) Surveyor's tape

Surveyor's tape is a special tape used to measure distances, made of non-stretch fabric that is durable and waterproof for tough outdoor working conditions. A typical surveyor's tape can be seen in **Error! Reference source not found.**.

Figure 1-1 Surveyor's tape.

The following are the various types of tapes

i. Cloth tape
ii. Metallic tape
iii. Steel tape
iv. Invar tape

Among the above, metallic tapes are widely used in surveying. A metallic tape is made of varnished strip of waterproof line interwoven with small brass, copper or bronze wires. These are light in weight and flexible and are made 2m, 5m 10m, 20m, 30m, and 50m.

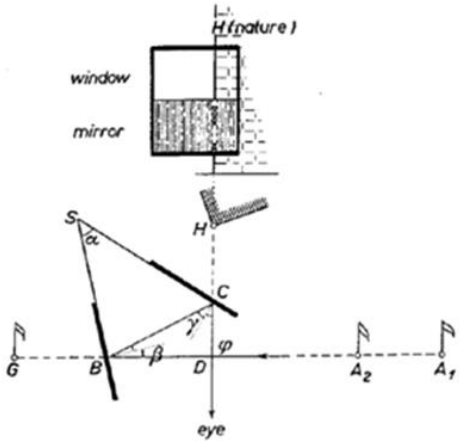
Surveyor's compass consists of a metal baseplate with two sighting wanes at the end. There is a compass box and two level vials that are perpendicular to each other. The compass is ready to use when the compass is set up and the vials are centered. If turning the compass box and the sighting vanes to sight along the desired line the magnetic bearing can be read.

Range pole or **lining rod** is usually made of wood, steel or aluminum and is marked with alternate red and white bands (see *Figure 1-2*) that can be used for rough measurements, to mark the line that being measured so that the tape's alignment can be maintained.



Figure 1-2. Range poles.

The angle mirror consists of two plane mirrors fixed at 45° to each other as seen Figure 1-3 Operating principle of an angle mirror. The images of point A_1 and A_2 are reflected by the mirrors at B and C and deflected angle φ from its original direction.



$\alpha = 45°$
$\varphi = 2\beta + 2\gamma$
$\alpha = \beta + \gamma$
$\varphi = 2\alpha = 2 * 45^0 = 90^0$

Figure 1-3 Operating principle of an angle mirror.

Right angle prisms (see *Figure 1-4***)** are made of glass. Knowing the laws of reflection and total reflection respectively, the optical path of the rays can be easily derived.



Figure 1-4. Right angle prism.

Pentagon prism was originally designed as a quadrilateral prism, with the fifth side obtained by cutting one edge. The two longer edges are mirror coated. The ray path is illustrated in Figure 1-5. The pentagon prism has a similarity with the angle mirror as a deflection of 90º is achieved by the angle of 45° that the coated sides of prism (i.e. the two longer sides) produced. The prism has much larger field of view that the angle prism and the images are brighter. The right angle is constructed inside the prism. The housing of the prism is sat on the top of a handle which has a cut for the plumb bob.

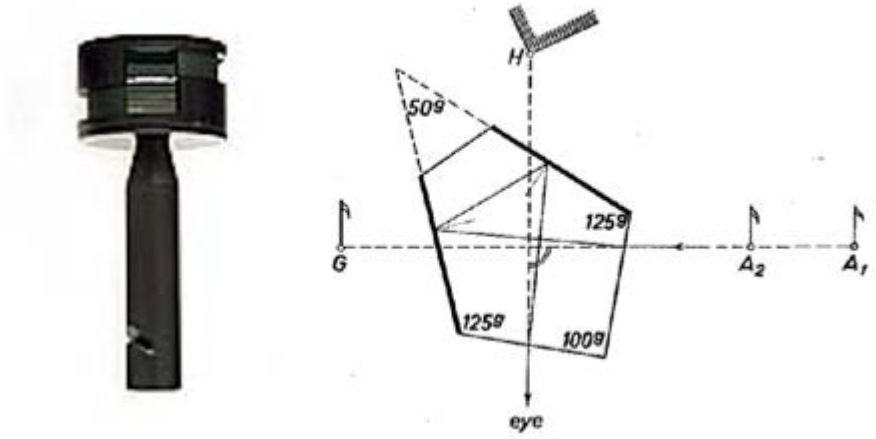


Figure 1-5 Pentagon Prism

1.4 Procedure

1.4.1 Staking of Straight Lines

A straight line may be staked in the field by using range poles. If there are no obstacles between the end points A and B of the line and distance AB is not too long, the staking of it is done by placing additional range poles at certain interval of say 30 to 50 m starting with the farthest point. Familiarize yourself with the method of staking straight lines by using range poles.

Staking of Straight Lines with Obstacles

If the end points of a straight line are not accessible (e.g. corners of a house) or not intervisible due to terrain then simultaneously lining by two operators should be applied (see *Figure 1-6* below).

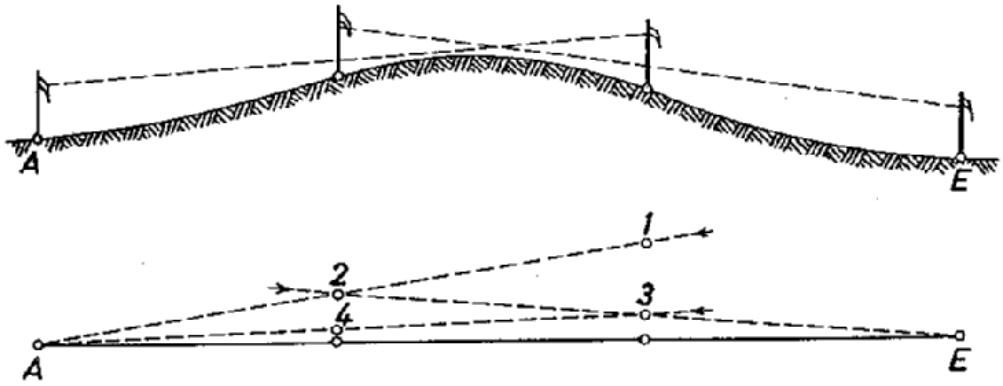


Figure 1-6. Simultaneous lining of range poles by two operators.

In case a house or other massive structure obscures the visibility between the end points then the staking may be done from a line parallel to the original one. Three different cases are presented in *Figure 1-7* below. Familiarize yourself with the three methods of staking straight lines with obstacles.

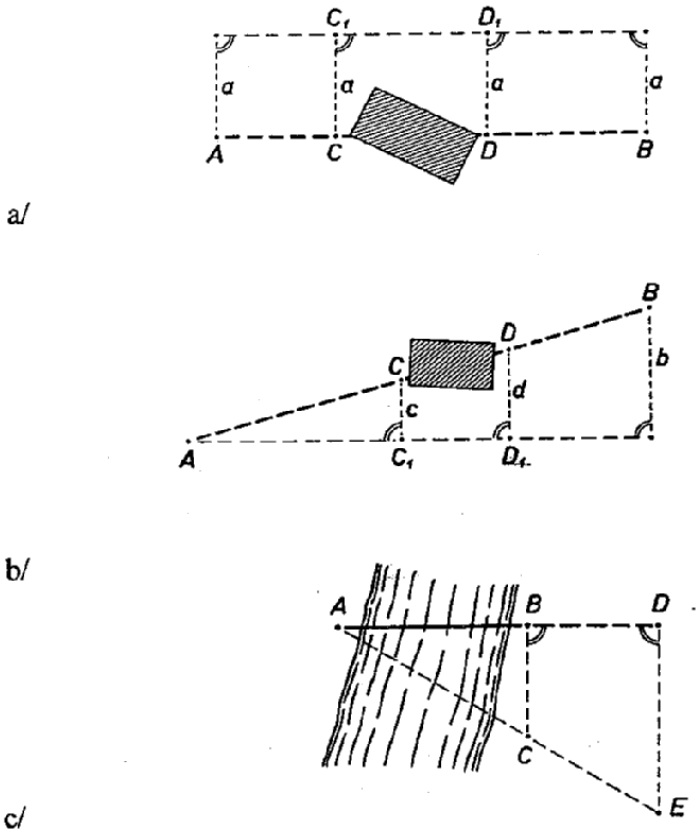


Figure 1-7. Staking when measuring line is obstructed: a) parallel staking; b) use of auxiliary base and c) crossing of a small river.

1.4.2 Staking of Fixed Angle

For a staking out of point at right angles to a reference line (measuring line), simple instrument like angle mirrors or different prisms are used.

Right Angle Measurement Using Square Prism

This instrument is used to measure a right angle from a point (C) outside a survey line to the survey line (AB) as seen in *Figure 1-8*.

→ First mount a ranging pole at A, B and C. The aim is to fix point D, so that CD is perpendicular to the survey line AB.

→ Go with the square prism should stand around D and move in the direction of AB and CD to see all the range poles in the prism mirror aligned vertically.

→ When all are aligned CD ⊥ AB, drop the plumb bob from the square prism to mark point D on the ground.

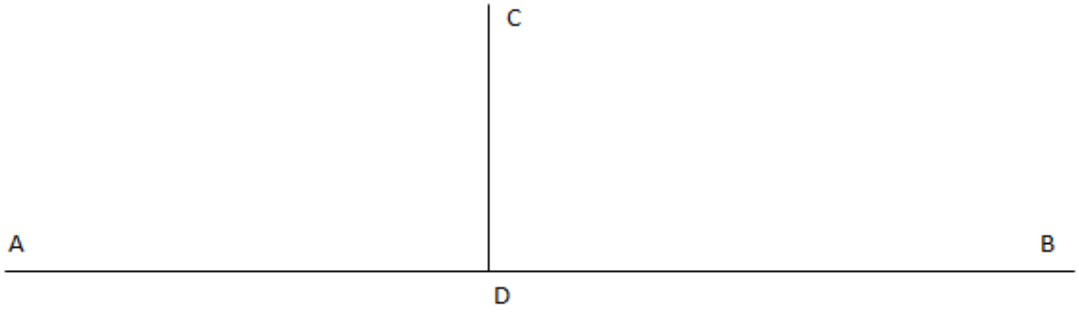


Figure 1-8. Staking out a point D at right angle to the survey line AB with square prism.

Erecting Perpendicular Line Using 3-4-5 Method

This method is used to set out a perpendicular line from a point (D) on the survey line (AB) to outside point (C) as seen in *Figure 1-9*. The steps are the following:

→ One person fixes 0 and 12 m mark of a tape at D.

→ Another person holds a mark 4 m from point D in the direction of the survey line.

→ The third person stretches the tape by holding the 9 m mark forming a right angle triangle. This gives a point C which form right angle at D to the survey line AB.

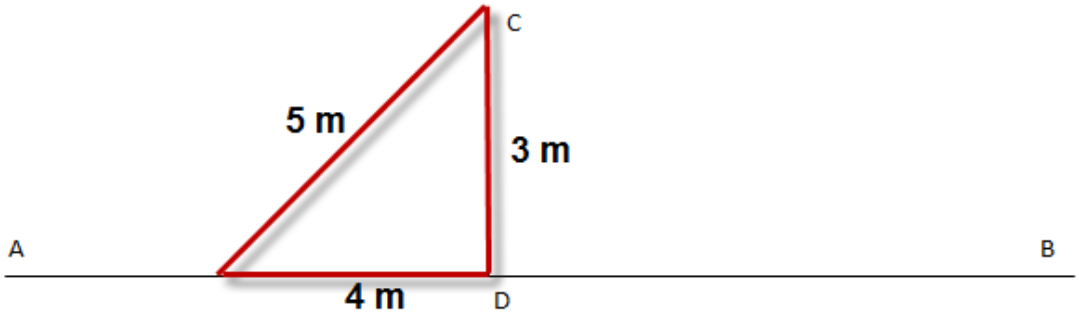


Figure 1-9. The 3-4-5 method.

1.4.3 Detail Survey

Detail surveys may be conducted either by aerial photogrammetry or by ground survey methods. Our interest is with the latter. These surveys are based on a control network. An area may be properly surveyed if only sufficient measurements, like angles and distances, are taken to permit plotting of all necessary points onto the plan/map. Small areas can be surveyed by tape and a pentagon prism (a rectangle device). Mainly two methods are used:

- The orthogonal (offset) method (see *Figure 1-10*) and
- The polar method.

Occasionally a tie-in method is used as a supplement to the orthogonal method. This method utilizes distance measurements only (see *Figure 1-11*). Familiarize yourself with the methods of detail survey (orthogonal, polar and tie-in).

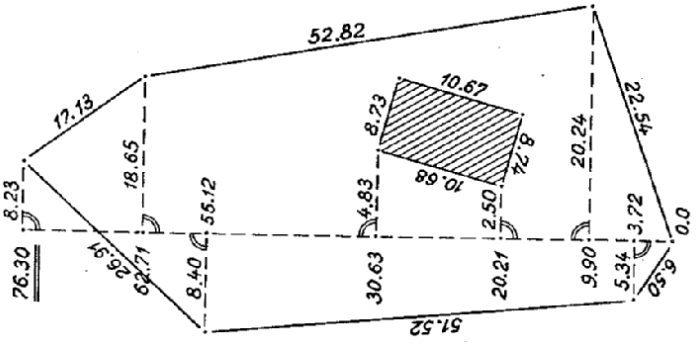


Figure 1-10. Orthogonal method used to survey a plot.

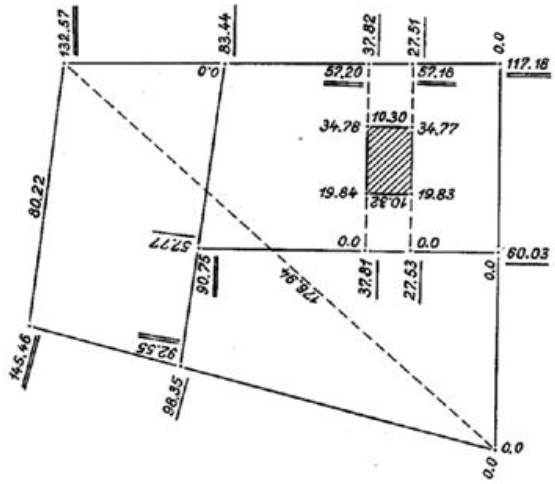


Figure 1-11. A typical tie-in survey of a plot.

1.5 Report Requirement

The report shall cover all the methods performed. It should contain all given and measured data with finalized drawing and computation. Introduction, objective, used instrument(s), analysis on the field challenges as well as recommendation and conclusion on the individual work.

Test 2

2 Introduction to Leveling

2.1 Objective of the Exercise

The objective of this exercise is to make you familiar with the leveling instrument and the commonly used leveling techniques, error detection and adjustment techniques. You are going to learn how to check and distribute errors in leveling when measuring in a closing loop, when you return to the same known point where you started, being able to check if there is any misclosure. In theory you should get back the same height. In practice you are likely to accumulate errors, and these errors must be distributed.

2.2 Theory

In surveying, three basic quantities are measured – heights, angles and distances – levelling is the name given to one of the methods available for determining heights. When levelling, it is possible to measure heights within a few millimetres and this order of precision is more than adequate for height measurement on the majority of civil engineering projects. As well as levelling, it is worth noting that heights can also be measured by using total stations, handheld laser distance meters and GNSS receivers – these are described in subsequent exercises of these manuals. In comparison to these, levelling offers a versatile yet simple, accurate and inexpensive field procedure for measuring heights and this is the reason for its continued use on construction sites in competition with other methods.

All methods of height measurement determine the heights of points above (or below) an agreed datum. On site or in the office, surveyors, builders and engineers all use, on a daily basis, horizontal and vertical datum as references for all types of measurement including levelling.

The heights of points relative to a datum are known as *reduced levels* (RLs). Any permanent reference point which has an arbitrary height assigned to it or has had its height accurately determined by levelling is known as a *bench mark*. For most surveys and construction work, it is usual to establish the heights of several marks throughout a site and if these have heights based on an arbitrary datum, they are known as *Temporary Bench Marks* (*TBM*s).

Definitions of terms

- **Datum: - A** datum is any reference surface to which the elevation (vertical distance) of a points are referred. The most commonly used datum is that of mean sea level.
- **Elevation: -** Elevation is the vertical distance of a points above or below on assumed datum (level surface).
- **Leveling: -** The process or methods of determining the vertical distance of a points relative to on assumed level surface.
- **Level line: -** is the surface of which it has a constant height relative to mean sea level.
- **Horizontal line:-** this is a line which is tangential to the level line or a line which is normal to direction of gravitas
- **Bench Mark (BM):-** are permanent reference points or marks at which their elevation (reduced level) has been accurately determined by leveling from other permanent BM.
- **Reduced level (RL):-** is the height above or below a reference datum- similar to elevation.
- **Temporary bench mark (TBM):-** are marks let up on stable points near construction sites which all leveling operation on that particular site will be referred.
- **Back sight (BS):-** is the staff reading taken on points of known elevation as a BM or a turning points.
- **Fore Sights (FS):-** is the staff reading on points whose elevation is to be determined as a turning points. It is the last staff reading denoting the shifting of the instruments.
- **Intermediate sights (IS):-** any other staff reading taken on a points at unknown elevation from the same set up of the level. All sights b/n BS & FS are IS.
- **Turning points (TP):-** is a point denoting the shifting at the level. It is the point on which the back a fore sight are taken.
- **Station: -** is a points of which whose elevation is to be determined.
- **Height of instruments: - is** the elevation of plane of collimation (plane of sight) where the instruments is correctly leveled.

2.3 Instruments Used

The instruments used for this exercise is leveling instrument, tripod, leveling rod, and surveyor's tape.

A leveling instrument is a telescope with magnifying effect that is used to read graduated rods (as seen in *Figure 11*) that are held on fixed points. The parts of the instrument are visualized in *Figure 2-1* below. The line of sight for the leveling instrument is always horizontal, unless there is collimation error.



Figure 2-1. The leveling instrument and its parts.

Leveling Staff: Levelling staff is wooden or aluminium made rectangular rod, graduated as to easy the reading. The levelling staffs are available in both metric and imperial graduation and sometimes even single staff contains both graduation on two sides. The level staff helps to determine the height difference between two points on a survey. The sliding joints of a levelling staff help to adjust the length of staff according to the need of use. A staff can be divided into two depending on its use as, self-reading staff and target staff. The only difference between self-reading staff and target staff is an attachment of metallic circular plate with a vernier, which called as target. A levelling staff (see is the equivalent of a long ruler and it enables distances to be measured vertically from the horizontal plane established by a level to points where heights are required. Many types of staff are in current use and these can have lengths of up to 5 m. The staff is usually telescopic but can be socketed in as many as five sections for ease of carrying and use and it is made of aluminum or non-conductive fiberglass.

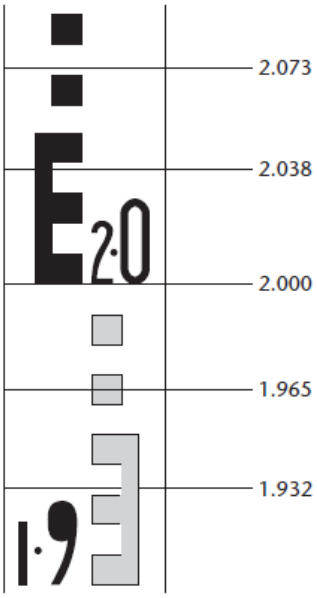


Figure 2-2 Leveling staff with example reading

2.4 Procedure

2.4.1 Level the Instrument

When the leveling instrument is set up on the tripod, the first step is always to level the instrument, by manipulating the screws on the tribrach. One quick leveling method is shown in Figure 2-3. If you assume that there is a line connecting two of the three screws you can adjust the position of the bubble along this imaginary line by turning these two screws simultaneously towards or away from each other. When the bubble is centered in that direction, you make the final adjustment by turning ONLY the third screw. In this case if you imagine that there is another line from the third screw that is perpendicular to the first imaginary line, the bubble will run along this when adjusted. This method can be applied for all kinds of instruments to be leveled.



Figure 2-3 Leveling the instrument with the screws of the tribrach.

2.4.2 Principle of Leveling

The principle of determining heights with leveling instrument and a leveling rod is shown in Figure 2-4 below.

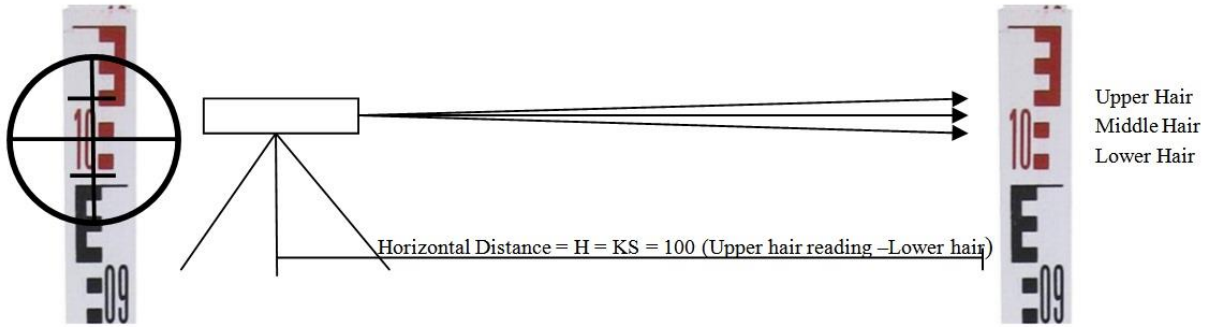


Figure 2-4 Leveling instrument with rod. The arrows from the leveling instrument show the readings for the upper, middle and lower crosshair. The first image shows the view seen through the ocular with the three crosshairs, aiming to the rod.

The basic principle of leveling can be understood from
Figure 2-5, where point A is a benchmark with the known elevation of 1800 m. The elevations at point B, C and D can be calculated with *Equation 1*.

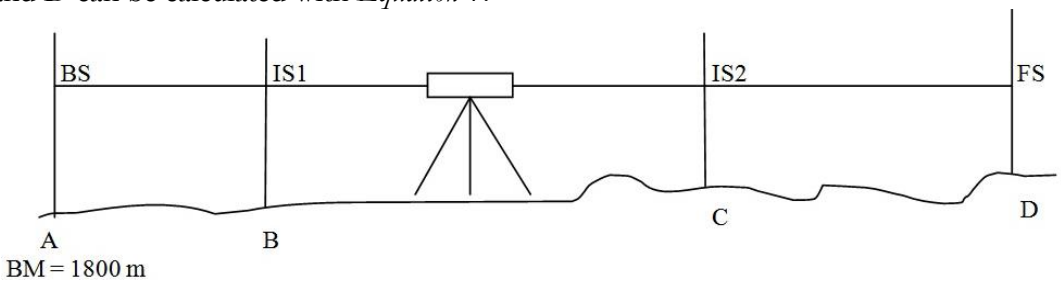


Figure 2-5. An example of a leveling assignment. The elevation of A is known, 1800 m. BS stands for back sight, FS for foresight while IS is an abbreviation intermediate sight.

Example of how to calculate the elevation of points B, D and C:

Elevation of B = Elevation of A + BS – IS1
Elevation of C = Elevation of A + BS – IS2
Elevation of D = Elevation of A + BS – FS

A typical error for leveling is the collimation error that can be eliminated with the *two peg test*. Two different cases can be distinguished for the collimation error:

- When the distance from the instrument and to the foresight, respectively to the back sight are approximately equal. An example of this case can be seen in
- *Figure* 2-6 below.

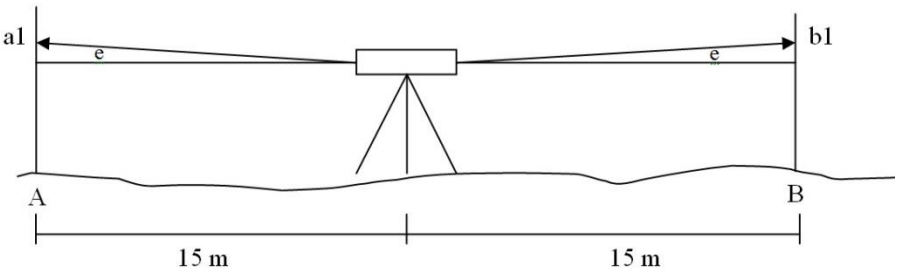


Figure 2-6. Leveling instrument with equal distance to the foresight and backsight.

$$\Delta H_{AB} = b1 - a1 \qquad (1)$$

- When the distance between the leveling instrument and the foresight, respectively backsight are not equal (see
- **Figure** 2-7).

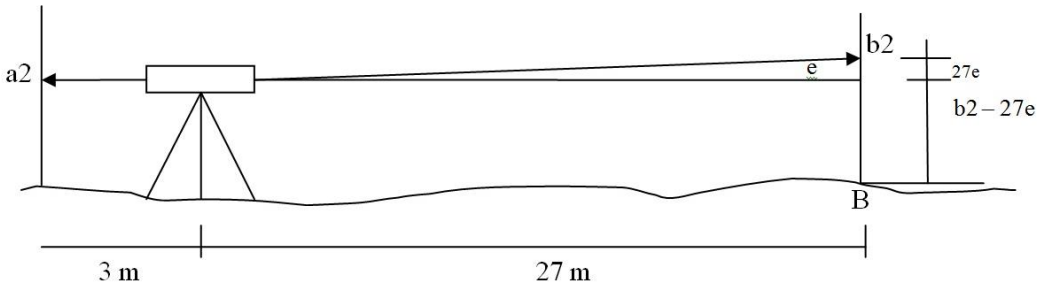


Figure 2-7. Leveling instrument with unbalanced distance to the foresight and backsight.

$$\Delta H_{AB} = (b2 - 27e) - a2 \qquad (2)$$

Equating equation *(1)* and *(2)* we get (b2 – 27e) – a2 = b1– a1, from where *e* can be expressed as the following

$$e = [(b2 - a2) - (b1 - a1)] / 27$$

If e is positive it means the instrument is inclines up so that we should subtract from the staff reading according to the distance from the instrument to the staff station, and vice versa.

2.4.3 Closing Loop

Inside your campus chose a route for the leveling loop starting with a BM with known elevation. Starting back sight as one of your BM and follow the leveling path to finish the leveling at the starting point as foresight. Calculate the corrected reduced level for each station using the format as Table 2-1.

Note: Choose the instrument station so that the distance between instrument station and the back sight approximately ***equal to distance*** *between the instrument station and the foresight.*

2.5 Report Requirement

After completing this exercise you shall prepare a report that includes the following:

- A short and concise introduction about leveling instrument and its use
- Computation of reduced level
- Closing loop, checking the presence of error, if any compute and correct the reduced levels of every station except the benchmark
- Explanation to the calculations and a table with readings and corrections.

2.7 Data Collection Format

Table 2-1 Table for leveling readings and calculations using Rise and Fall Method

Point No	Distance	Reading			Difference in Elevation		Reduced Level	Correction applied	Correct RL	Remark
		BS	IS	FS	Rise	Fall				

Table 2-2 Table for leveling readings and calculations using Rise and Fall Method

Point No	Distance	Reading			Instrument Height	Reduced Level	Correction Applied	Correct RL	Remark
		BS	IS	FS					

Exercise

3

3 Angle Measurement by Theodolite/Total Station

3.1 Objective of the Exercise

In this exercise you will familiarize yourself how to measure horizontal and vertical angle, calculate horizontal and vertical angle error and correct the angles for the errors.

3.2 Theory

Theodolites are precision instruments and are either electronic (capable of displaying angle readings automatically) or optical, which need to be read manually. Many different theodolites are available for measuring angles and they are often classified according to the smallest reading that can be taken with a particular instrument – this can vary from 1' to 0.1"

Total station instruments (also sometimes called electronic tacheometers) combine an EDM instrument, an electronic digital theodolite, and a computer in one unit. These devices automatically observe horizontal and zenith (or altitude) angles, as well as distances, and transmit the results in real time to a built-in computer. The horizontal and zenith (or altitude) angle and slope distance can be displayed, and then upon keyboard commands, horizontal and vertical distance components can be instantaneously computed from these data and displayed. If the instrument is oriented in direction, and the coordinates of the occupied station are input to the system, the coordinates of any point sighted can be immediately obtained. These data can all be stored within the instrument, or in a data collector, thereby eliminating manual recording.

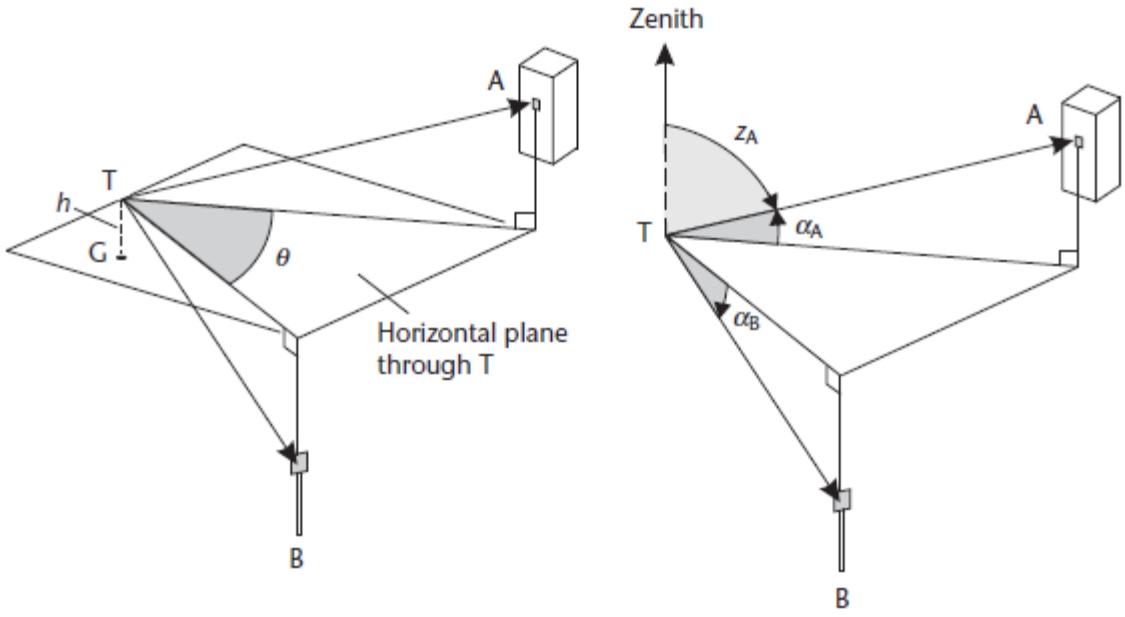


Figure 3-1 Horizontal, vertical and zenith angles

Horizontal angle is an angle between two lines in the horizontal plane (see *Figure 3-2*).

Vertical angle is an angle between two lines in the vertical plane (see *Figure 3-2*).

Azimuth is the direction angle within the plane of the horizon, counted clockwise from the North.

Zenith angle is the angle measured from the vertical (zenith direction) to a point.

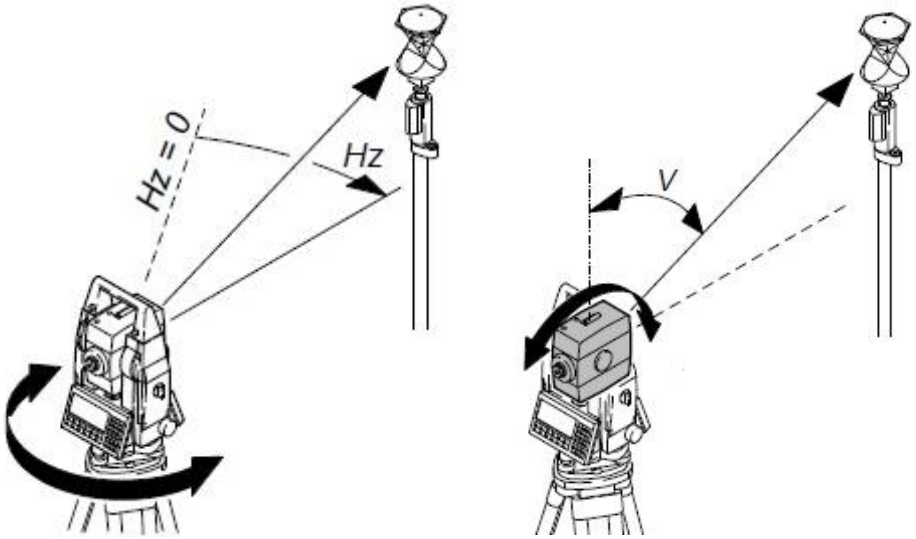


Figure 3-2. Total station and reflector, showing how to turn the instrument to measure horizontal, respectively

3.3 Instruments Used

The instruments needed for this exercise are total station, which's parts can be seen below in *Figure 3-3,* tripod and prism, which can be seen in *Figure 3-4* below.

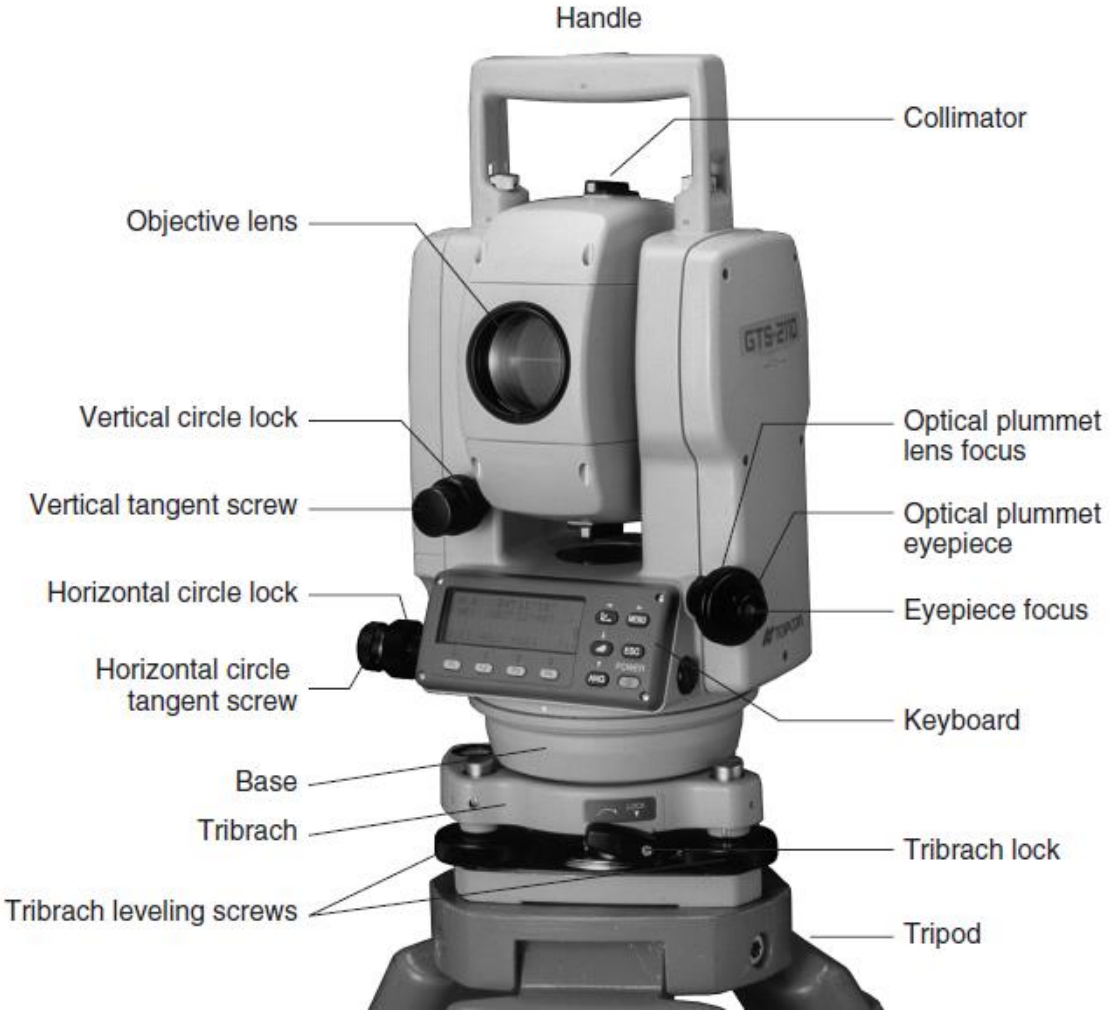


Figure 3-3. The parts of the total station.

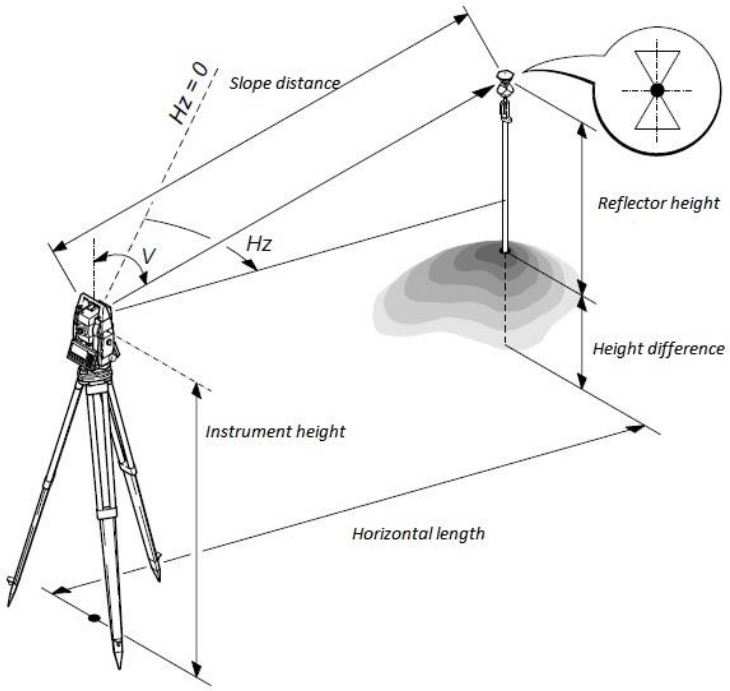


Figure 3-4. A total station mounted on a tripod aiming to a reflector.

3.4 Procedure

Horizontal and Vertical Angle Measurements

Set up the total station at point S (see *Figure 3-5*) and go through the leveling, configuration etc steps as described before and start up the surveying programme and press **Cont**.

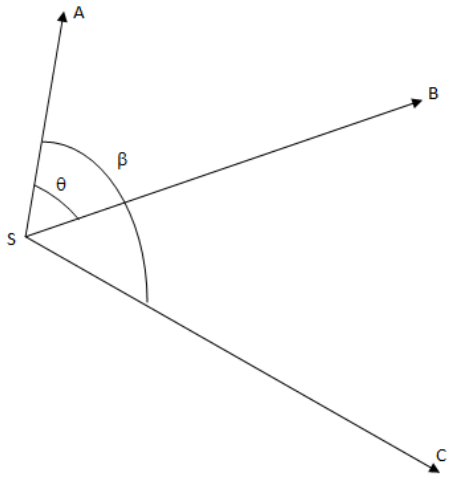


Figure 3-5. **Example, total station at point S, measuring at points A, B and C.**

→ Aim at A in face left mode and record the horizontal and vertical direction (visualization of these angles can be seen in *Figure 3-2*). Use Table 3-1 and Table 3-2 to record the results.
→ Aim at B in face left mode and record the horizontal and vertical direction
→ Aim at C in face left mode and record the horizontal and vertical direction
→ Turn the telescope 180° and the total station 180° to change the instrument to facer right position at C and record the horizontal and vertical direction, than repeat the same procedure at point B and then finally at point A
→ Now calculate the vertical (***Note: use zenith angle in this exercise!)*** and horizontal angles as in the example below.

3.5 Calculation

Example of Horizontal Angle Calculation
FR - FL = 180°
SIMPLE MEAN = (FR + FL ± 180°) / 2
IF FL > 180 ° THEN IT IS +
IF FL < 180° THEN IT IS -

Station	POINT	Face		Simple Mean	Reduced Mean	Total Mean	Diagram
		Left	Right				
S	A	00°03'50"	180°04'30"	00°04'10"	00°00'00"		
	B	17°22'10"	197°'23'10"	17°22'40"	17°18'30"		
	C	83°58'50"	264°00'00"	83°59'25"	83°55'15"		

Example of Vertical Angle Calculation

REDUCED FL = 90° - FFL
REDUCED FR = FR - 270°
MEAN = (RFL + RFR) / 2

STATION	POINT	FACE		REDUCED FACE LEFT (RFL)	REDUCED FACE RIGHT (RFR)	MEAN
		LEFT	RIGHT			
S	1	88°10'30"	271°51'20"	01°49'30"	01°51'20"	01°50'25"
	2	89°34'50"	270°'27'30"	00°25'10"	00°27'30"	00°26'20"
	3	92°44'10"	267°13'40"	02°44'10"	02°46'20"	02°45'15"

.

3.6 Report Requirement

The report shall be submitted containing description of the instrument and measurement method, together with a produced map of the surveyed area and a record of measurement data.

3.7 Tables and Notes

Table 3-1 Data recording format for horizontal angle

Instrument Station	Target Point	Face Left			Face Right			Simple Mean			Reduced Mean			Total Mean		
		°	‘	”	°	‘	”	°	‘	”	°	‘	”	°	‘	”

Table 3-2 Data recording format for vertical angle

Instrument Station	Target Point	Face Left			Face Right			Reduced Face Left			Reduced Face Right			Mean		
		°	‘	”	°	‘	”	°	‘	”	°	‘	”	°	‘	”

Exercise

4

4 Distance Measurement by Theodolite and EDM

4.1 Objective of the Exercise

In this exercise you will familiarize yourself how to measure distance by optical and EDM method using Theodolite and Total Station respectively.

4.2 Theory

4.2.1 Optical measurement method

Optical measurement method is also known as tacheometry method or stadia method. Optical measurement is inspection of measurement with the help of telescope and then calculations are solved using triangulation method to find out the distance. The instrument tacheometer is considered as important in this method. Other than these, *level staff* which is a graduated wooden or aluminium rod needed to carry out optical measurement. Before the electronic distant measurement method this method is more famous and accurate for a particular distance. This is a suitable method to survey a hill country and counter lines.

Tacheometer is an instrument used to measure the vertical and horizontal distance between two points. Tacheometer is more similar to theodolite but an optical lens is available in a telescope and two stadia line are available other than central cross hair. The telescope used to measure the distance between two points using the level rod. Then the distance between instrument and levelling staff is obtained by multiplying levelling staff interval by a constant. The common equation to obtain the distance, d between the instrument and levelling staff is,

$$d = k \times s + c$$

Here, s is the levelling staff interval (i.e. upper hair reading minus lower hair reading); k and c are multiplicative and additive constants. In most of the instrument the k is 100 and c is 0.

Horizontal Line of Sight

When the line of sight is horizontal (e.g. level instrument), the distance between the instrument and the staff is calculated by multiplying the stadia interval (i.e. upper hair reading – lower hair reading) with instrument constant (usually 100).

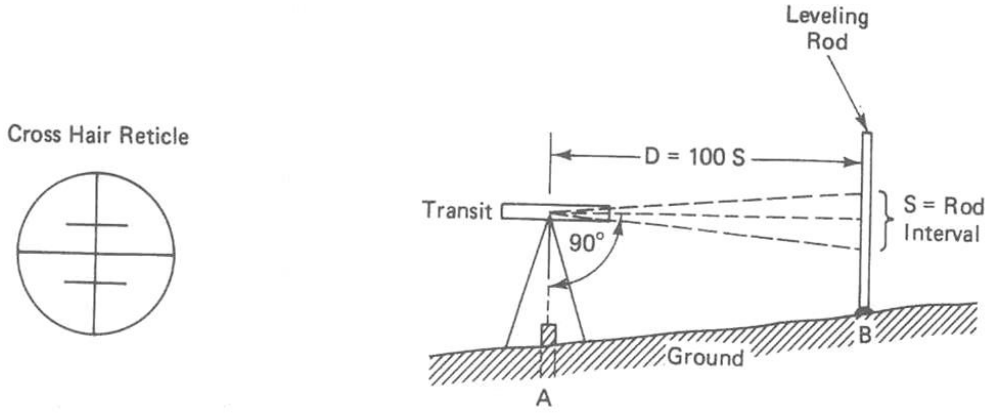


Figure 4-1 Optical distance measurement for horizontal line of sight

Inclined line of sight

When the line of sight is inclined (for Theodolite instruments), we can apply the following formula to calculate the slant, vertical, and horizontal distances.

Slant distance , $D = 100Scos\theta$
Horizontal Distance, $H = Dcos\theta = 100Scos^2\vartheta$
Vertical distance, $V = Dsin\theta = 100Scos\vartheta sin\theta$

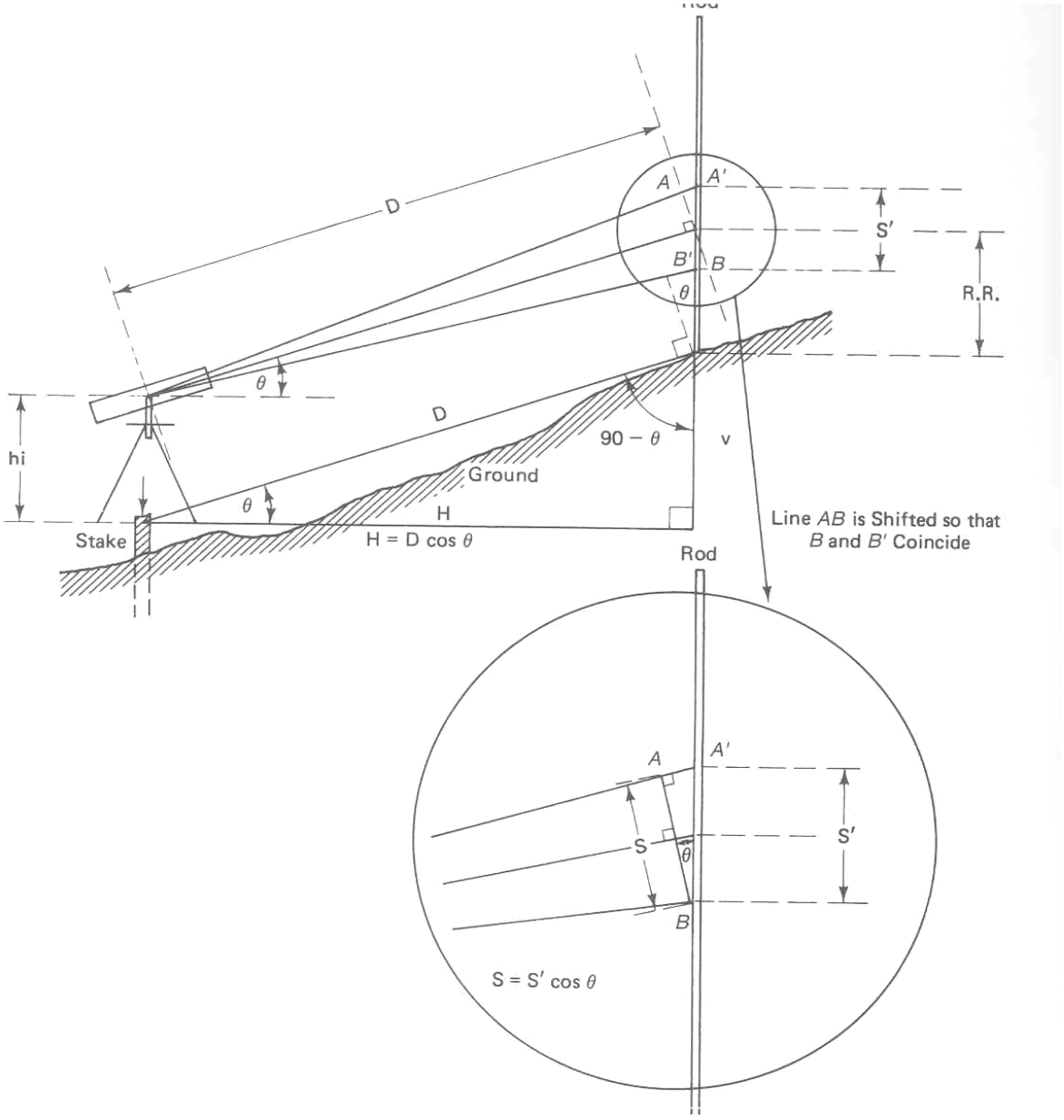


Figure 4-2 Optical distance measurement for inclined line of sight

4.2.2 Electronic Distance Measurement (EDM)

When a distance is measured with a total station, an electromagnetic wave or pulse is used for the measurement – this is propagated through the atmosphere from instrument to reflector or target and back during a measurement. Distances are measured by one of two methods: the phase measurement method, which uses continuous electromagnetic waves, or the timed pulse method, in which pulses of laser radiation are used.

The timed-pulse method is a much simpler approach to distance measurement compared with the phase measurement method, which was originally developed about 50 years ago at a time when electromagnetic distance measurement was first invented. Although methods based on transit times

were considered at that time, it was not possible to measure these with sufficient accuracy. Consequently, the phase measurement method was adopted, but many total stations now use the timed-pulse method.

Error in Electronic Distance Measurement Instruments

Personal Errors

- Inaccuracy in initial setups of EDMs and the reflectors over the preferred stations
- Instrument and reflector measurements going wrong
- Atmospheric pressures and temperature determination errors

Instrumental Errors

- Calibration errors
- Chances of getting maladjusted time to time generating frequent errors
- Errors shown by the reflectors

Natural Errors

- Atmospheric variations in temperature, pressure as well as humidity. Micro wave EDM instruments are more susceptible to these.
- Multiple refraction of the signals.

The advantage of using EDM instruments is the speed and accuracy in measurement. Several obstacles to chaining are automatically overcome when these instruments are used.

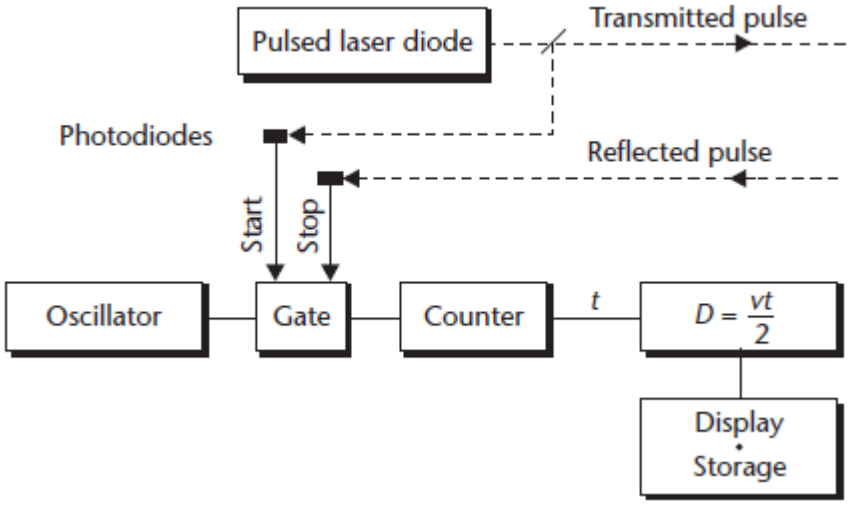


Figure 4-3 Timed-pulse distance measurement

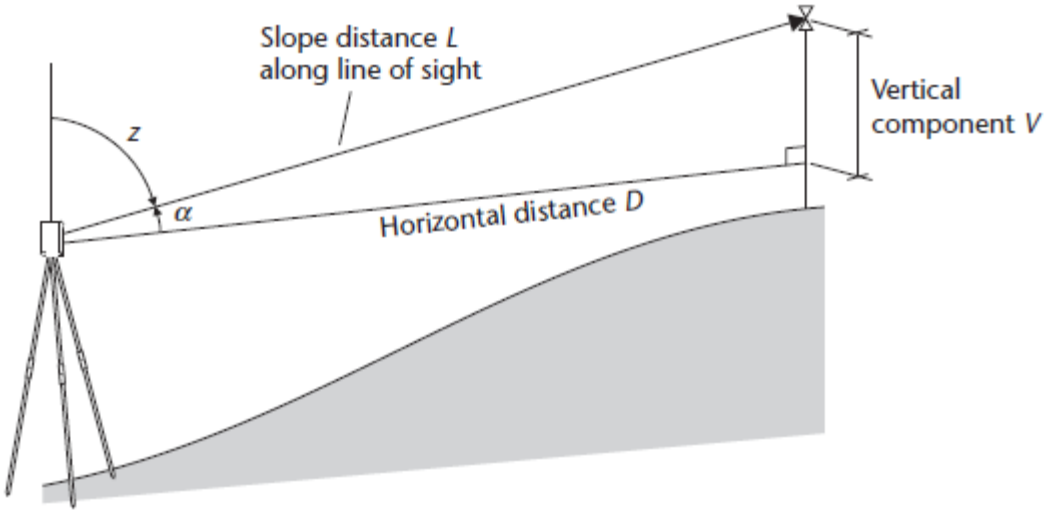


Figure 4-4 EDM measurement

Horizontal distance, D = L cos α = L sin z
Vertical component, V = L sin α = L cos z
where α is the vertical angle and z is the zenith angle.

4.3 Instruments Used

In this exercise, we used instruments like theodolite, total station, level staff and reflectors.



Figure 4-5 Parts of Digital Theodolite

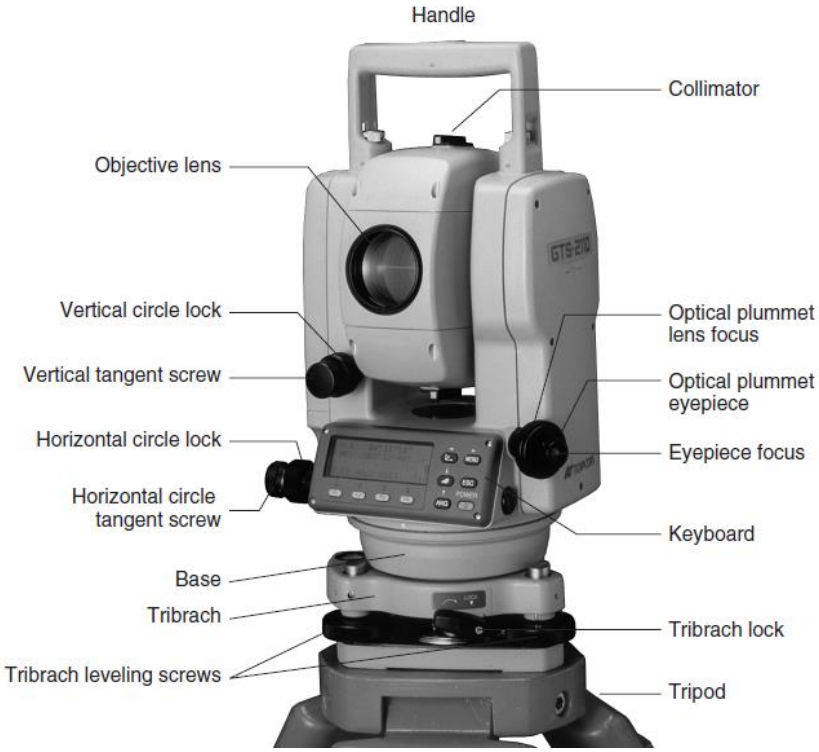


Figure 4-6 Parts of Total Stations

4.4 Procedure

Before proceeding to measurement, choose three points A, B, and C. At point A we set the instrument and pick higher point B and lower point C as a target.

Measurement of Distance with Theodolite

1. Set the theodolite instrument at A (see Figure 4-7) and level and center the instrument
2. Measure the height of instrument using Tape
3. Target the Theodolite to point B.
4. Read the upper hair, middle hair and lower hair on the staff at B and the zenith angle.
5. Do the same from 3-4 for target at C.

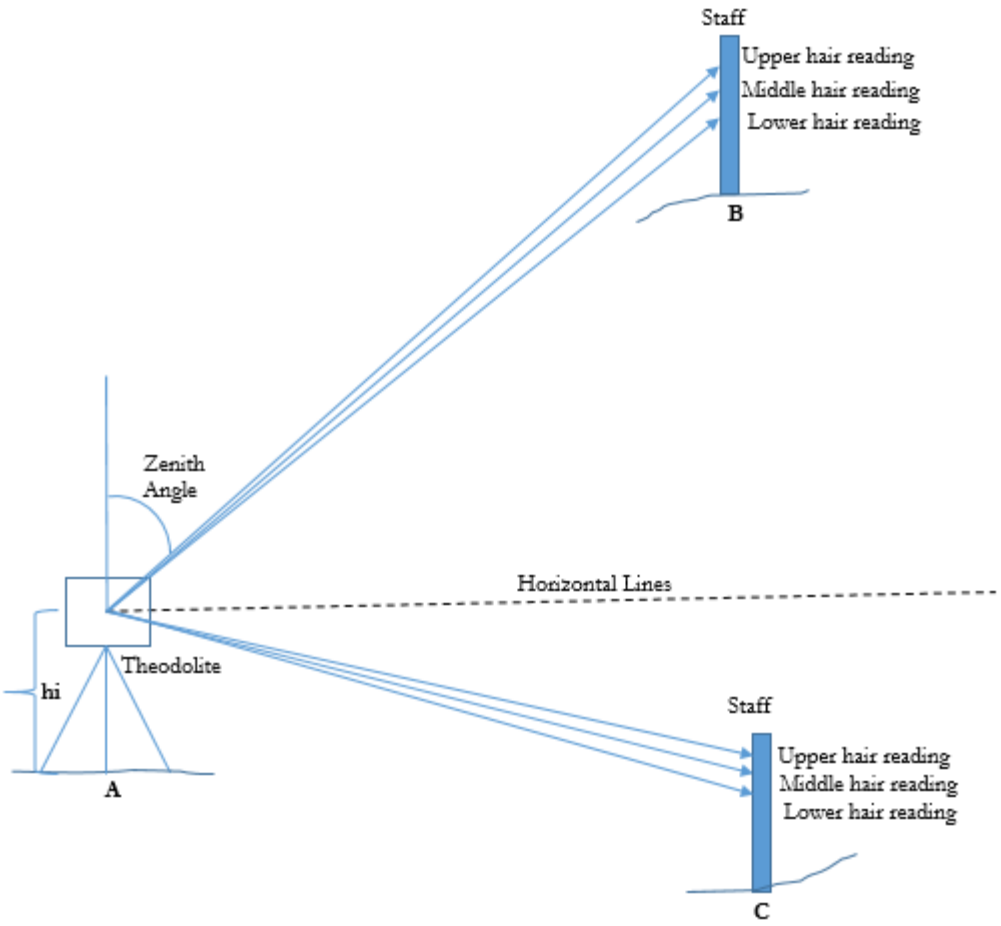


Figure 4-7 Optical distance measurement

Measurement of distance with EDM and a Reflector

1. Set the Total Station instrument at A (see Figure 4-8) and level and center the instrument
2. Measure the height of instrument using Tape
3. Target the Total Station to point B.
4. Measure the distance using the reflector at B and read the zenith angle at B
5. Do the same from 3-4 for target at C.

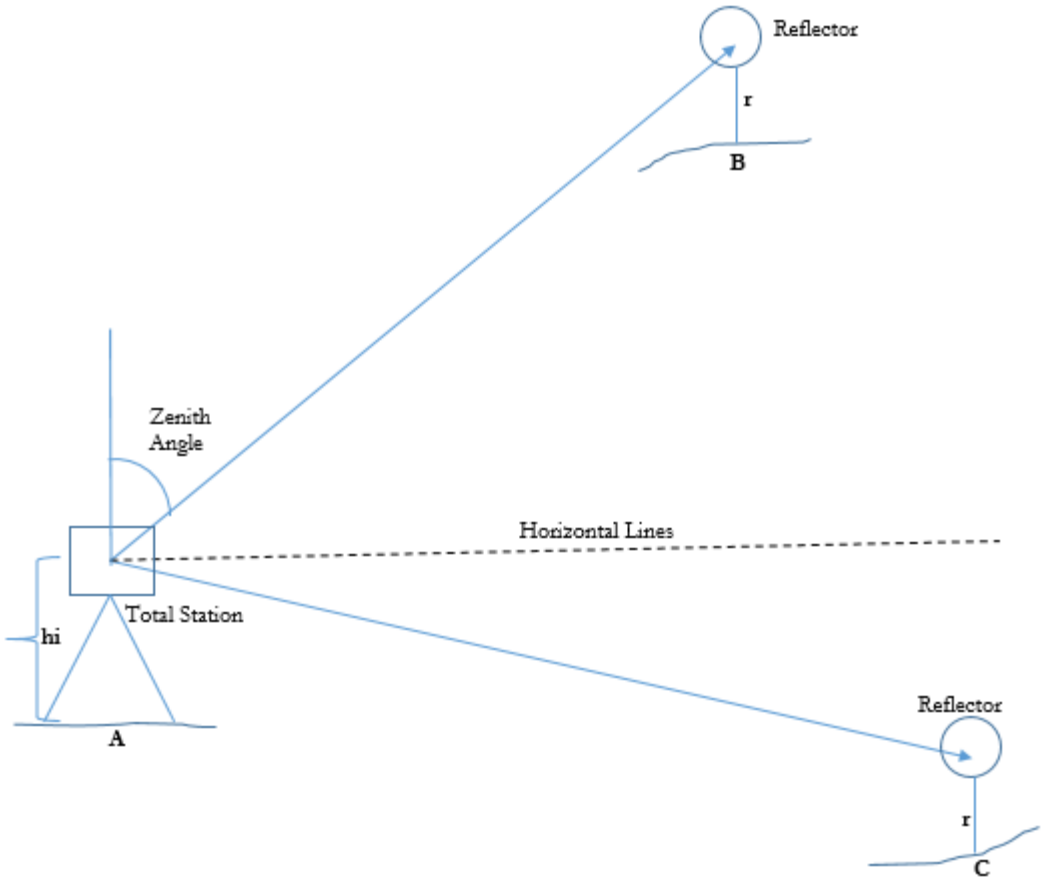


Figure 4-8 Electronic distance measurement

4.5 Report Requirement

The report should include introduction, theory, data collected, and computations. The computations should include calculations of horizontal and vertical distances using optical and EDM methods.

Exercise

5

5 Trigonometric Leveling

5.1 Objective of the Exercise

This exercise will help you understand how height of towers/buildings are determined using Trigonometric leveling.

5.2 Theory

The raw data measured by a total station is horizontal angle, zenith (or vertical) angle and slope distance all of which are usually converted into the three dimensional coordinates of the position of the reflector. This section gives details of how the height of the point sighted is obtained – the procedures for obtaining the horizontal position are given in traversing Exercise 7.

To illustrate how a total station is used to determine heights over short distances, consider Figure 5-1, which shows two points A and B. A total station is levelled and centered at A and a reflector is set up at B. The slope distance L and zenith angle z (or vertical angle *a*) between A and B are measured together with the height *hi* of the total station above A (height of instrument) and the height hr of the reflector above B (height of reflector). If the height of A is known (HA), the height of B (HB) is given by

$$HB = HA + hi + VAB - hr$$

The vertical component VAB is obtained from

$$VAB = L \cos z = L \sin \alpha$$

and this will be positive if the telescope is tilted above the horizontal through the total station (z less than 90° or positive vertical angle) and will be negative if it is below the horizontal (z greater than 90° or negative vertical angle). This is known as trigonometrical heighting and is the basis for all height measurement with a total station. It gives heights of sufficient accuracy for most detail

surveys and for some setting out up to a distance of about a hundred meters. It is also used in other application programs by a total station when heights are required for three-dimensional traverses, when performing a resection with elevations and for remote height measurements.

For all trigonometrical heighting, it is important that the reflector is sighted accurately and that zenith (or vertical) angles are measured carefully, especially for traverses, resections and other control surveys. It is also important that the correct instrument and reflector heights are measured and entered into the instrument. When taking readings to a detail pole, the pole is often extended so that it is the same length as the height of the instrument – this simplifies any manual calculations of height since *hi* will equal *hr* to give *H*B = *H*A + *V*AB. However, there is a tendency for telescopic detail poles to shorten during a long set up and their lengths should be checked frequently. When a lot of points are to be measured from a single instrument position, it is better to use a pole fully extended or, better still, to use a fixed length pole – these are available in various lengths.

As well as being able to measure ground heights at a reflector, a total station can also be used in reflectorless mode to determine the heights of any point targeted, including inaccessible ones where it is not possible to locate a prism. In this case, there is no height of reflector to measure and *H*B = *H*A + *hi* + *V*AB.

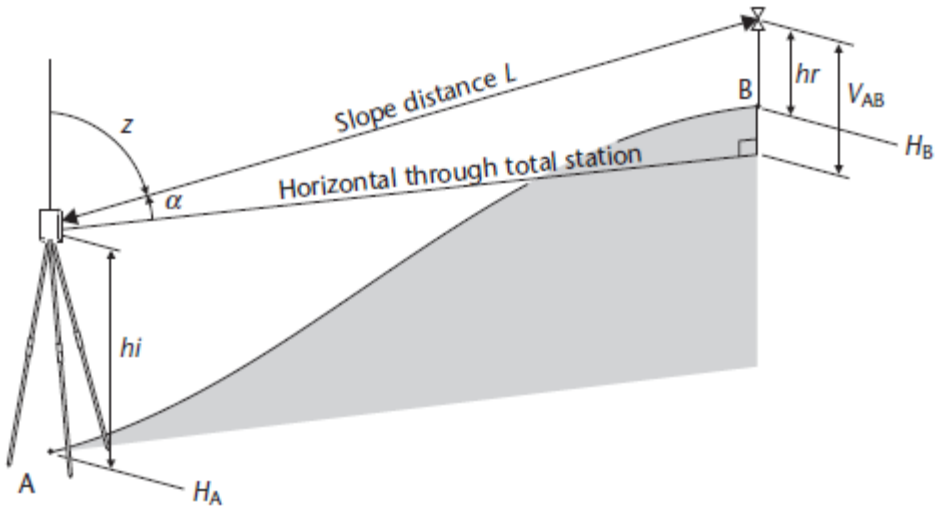


Figure 5-1 Trigonometric leveling

5.3 Instrument Used

In this exercise, Total Station, Tripod, Tape, and Reflector are used.

5.4 Procedure

Trigonometrical Levelling: Base Accessible

It is required to find the elevation (R.L.) of the top of a tower Q from the instrument station P as shown in Figure 5-2.

1. Set up theodolite at P and level it accurately with respect to the altitude bubble. See that the vertical circle reads 00 0' 0" when the line of sight is horizontal.
2. Direct the telescope towards Q and bisect it accurately, clamp both the plates. Read the zenith angle.
3. Plunge the telescope and sight to the same point Q in face right and take the zenith angle. Calculate the average of the vertical angles measured in both faces.
4. Measured horizontal distance at base of the tower by EDM method (D).
5. With the vertical circle set to zero reading and the altitude bubble in the centre of its run take the reading on the levelling staff kept at A.B.M. Let it be S.
6. Calculate the reduced level of Q by:

   R.L. of Q = R.L. of A.B.M. + S + h

   where h = D tan α

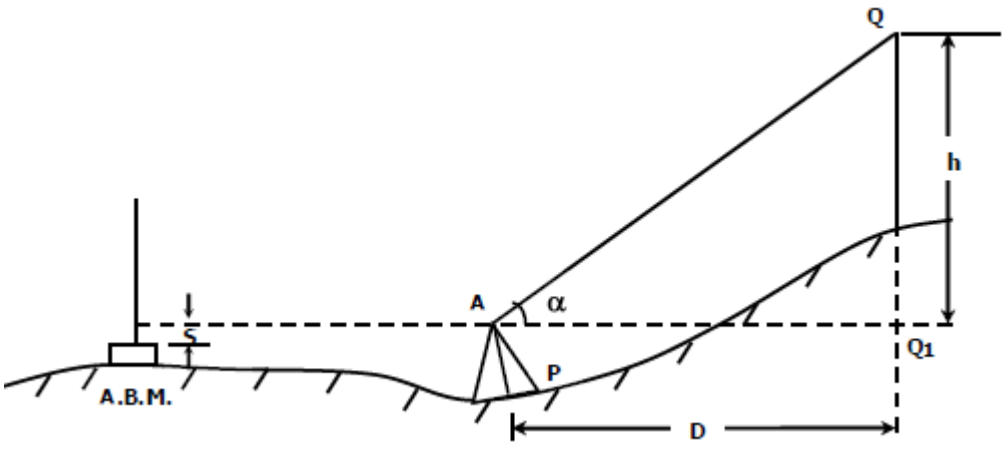


Figure 5-2 Trigonometric levelling when the base is accessible

5.5 Report Requirement

The report should include the collected data and computation of tower height and should presented in standard reporting format.

Exercise

6

6 Detail Measurement Using Total Station

6.1 Objective of the Exercise

The objective of this exercise is how the coordinates of the unknown points are determined using Total Station from known stations.

6.2 Theory

Coordinate measurements with total stations

The horizontal circle of a total station can be set to read whole-circle bearings directly using the following procedure. First, the coordinates of the instrument station are entered into the total station, followed by the coordinates of a reference station, as shown in Figure 6-1. Second, the reference station is sighted and an orientation program in the instrument is activated to carry out a rectangular → polar conversion and calculate the whole-circle bearing A from the instrument station to the reference station and to set the horizontal circle to display this bearing. This is known as horizontal circle orientation, and whatever direction the total station is pointed in after this, it will display the azimuth along that direction.

Having orientated the horizontal circle of a total station, the coordinates of other points can be determined fairly easily. In, a new point C is sighted and the total station will display the azimuth A to this (A_{AC}). The horizontal distance D (of AC) is measured and, using the coordinate measurement program, the instrument will now carry out a polar→rectangular conversion to calculate and display the coordinates of the new point. This can be extended to three dimensions if the height of the instrument station A and appropriate instrument and reflector heights are also entered into the total station, which will then display the height of the new point.

Azimuth of AB $= tan^{-1}\left(\frac{E_B - E_A}{N_B - N_A}\right)$

$$E_C = E_A + D\ Sin\ A_{AC}$$

$$N_C = N_A + D\ Cos\ A_{AC}$$

And from the concept of trigonometric surveying the height of C is determined using

$$H_C = elevation\ of\ station\ A + h\ \pm V - r$$

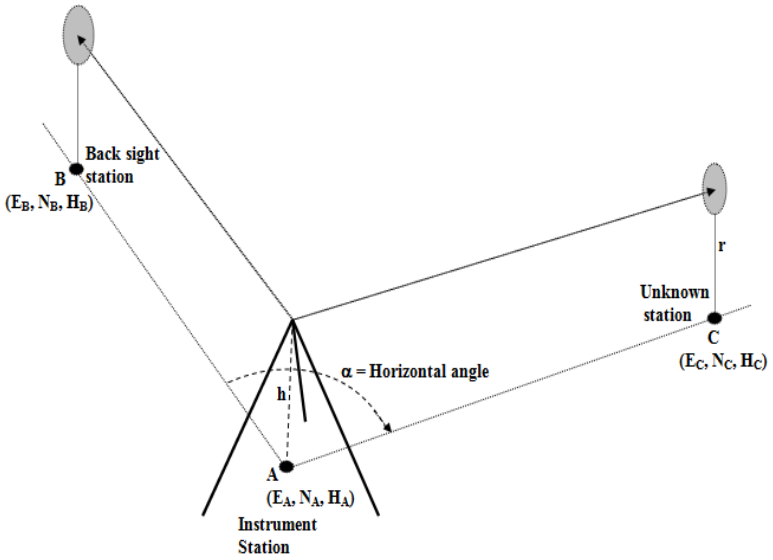


Figure 6-1 Principles of coordinate computation using Total Station

6.3 Instrument Used

In this exercise, Total Station, Reflector, Tripod, and Tape are used.

6.4 Procedure

1. Set, level, and center the instrument at A
2. Choose known backsight method and insert the coordinate of the instrument station and the backsight station coordinate
3. Target to point B
4. Set the instrument direction to azimuth calculated by the instrument (i.e. Azimuth of AB)

5. Rotate the instrument clockwise and point to the target of the unknown station and measure and record the data.

6. Repeat 5 to collect relevant data to get information for specific purpose.

7. When all the data you want is not visible from this station, change the instrument to another known station already established (for example by traversing) and repeat step 1 to 5 to get data for preparation of topographic map, design or other purpose.

6.5 Report Requirement

The report should include the concept of data collection using Total Station.

Exercise

7

7 Closed and Link Traverse

7.1 Objective of the Exercise

Traversing is one of the easiest and most popular methods of establishing control networks in surveying that can be used to make detail measurements, to map a plot, create a topographical or other maps using total station.

Traverse networks have the following advantages:

(1) Little reconnaissance is required compared with that needed for an interconnected network of points.
(2) Observations only involve three stations at a time so planning the task is simple.

7.2 Theory

A traverse is a means of providing horizontal control in which the rectangular coordinates of a series of control points located around a site are determined from a combination of angle and distance measurements. Control points in traversing are often called traverse stations. In Figure 7-1, a traverse has been run from existing control point A to unknown traverse stations 1–2–3 and then to existing control point X. This traverse opens along AB, fixes the positions of 1, 2 and 3 by measuring the angles and distances shown and then closes along XY. For this reason this traverse is called a *link* traverse (or sometimes a *connecting* or *closed-route* traverse). The origin and north direction of a link traverse are set by the coordinates of the opening control points A and B.

In Figure 7-2(a), a traverse starts at point A and returns to A via traverse stations 1, 2 and 3. In this case the traverse is called a *polygon* traverse (or sometimes a *loop* or *closed-ring* traverse) since it closes back on itself to form a polygon. The origin for this traverse is defined by point A which can be

an existing control point or a traverse station with an assumed position – an arbitrary bearing is assigned to one of the lines to define the orientation and north direction. Figure 7-2(b) shows another polygon traverse based on two existing control points in which the origin and orientation are defined by the coordinates of A and B.

Both the polygon and link traverses described here are known as *closed traverses*, since there is an external check on the observations and fieldwork as they both start and finish at known or assumed points.

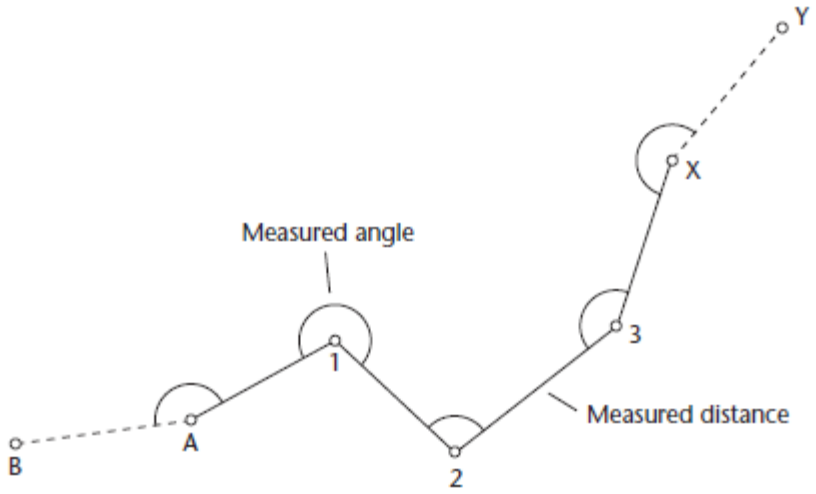


Figure 7-1 Link traverse

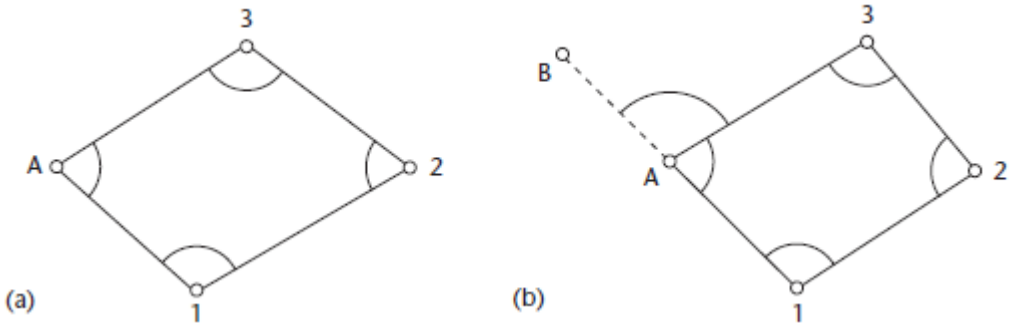


Figure 7-2 Polygon/Closed traverse

Traverse Adjustments

In theodolite traversing, angles and distance are measured and azimuth of one side of the polygon or direction must be known. The measured angles are adjusted to fulfill the following conditions:

$$\text{sum of } \textit{internal} \text{ angles} = (2n-4)\times 90°$$
$$\text{or sum of } \textit{external} \text{ angles} = (2n+4)\times 90°$$

where n is the number of angles measured.

When the bearings in a *link traverse* are calculated, an *initial back bearing* can usually be determined from known points at the start of the traverse and, to check the observed angles, a *final forward bearing* is computed from known points at the end of the traverse. The angular misclosure in a link traverse is found using

sum of angles = (final forward bearing – initial back bearing) + ($m \times 180°$)

In this equation, m is an integer, the value of which depends on the shape of the traverse. In most cases, m will be $(n-1)$, n or $(n+1)$, where n is the number of angles measured between the initial back bearing and final forward bearing.

From known azimuth, the azimuth of all sides are computed (see Figure 7-4 for closed traverse). The departure and latitude for each side are computed using:

Departure of the line = Distance of the line x Sin (Azimuth of the line)

Latitude of the line = Distance of the line x Cos (Azimuth of the line)

If the measurement is error-free, the summation of departure and latitude should give us zero. Practically these summations are different from zero due to error in the measurement of angles and distance. The summation of departure (e_E) and latitudes (e_N) in vector form is closure error (see Figure 7-3) of the traverse.

Bowditch method

The values of the adjustment found by this method are directly proportional to the length of the individual traverse lines. The method is best suited to taped traverse lines.

Adjustment to ΔE (or ΔN) for a traverse line

$$= \delta E \text{ (or } \delta N) = -e_E \text{ (or } -e_N) \times \frac{\text{length of traverse line}}{\text{total length of the traverse}}$$

After adjusting the departures and latitudes for closure, we can start calculating the coordinates of the unknown points from the known point of the traverse station. Similar approaches can be applied for the computation of coordinates in link traverse.

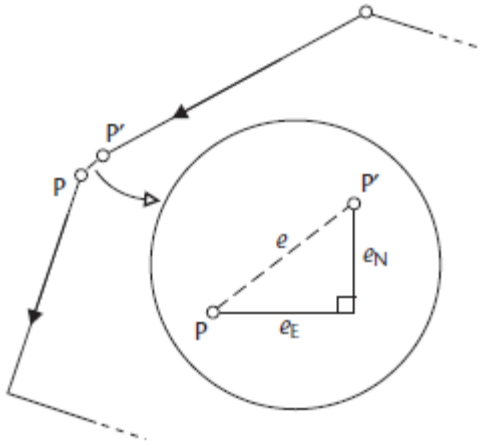
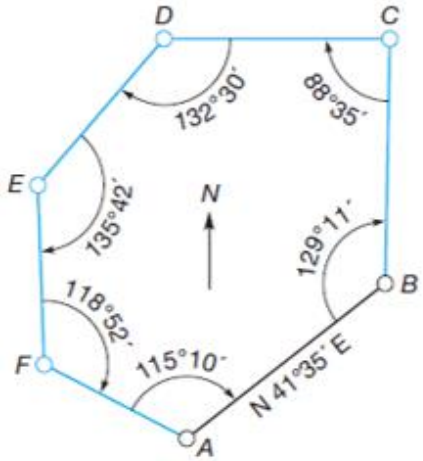


Figure 7-3 Traverse misclosure



41°35′ = AB	211°51′ = DE
+180°00′	−180°00′
221°35′ = BA	31°51′ = ED
+129°11′	+135°42′
350°46′ = BC	167°33′ = EF
−180°00′	+180°00′
170°46′ = CB	347°33′ = FE
+88°35′	+118°52′
259°21′ = CD	466°25′ − *360° = 106°25′ = FA
−180°00′	−180°00′
79°21′ = DC	286°25′ = AF
+132°30′	+115°10′
211°51′ = DE	401°35′ − *360° = 41°35′ = AB ✓

Figure 7-4 Example of azimuth computation for closed traverse

7.3 Instrument Used

The exercise can be carried out with the help of a total station, theodolite, compass, tripod, two prisms, a pocket tape and paint (or drilling machine).

7.4 Procedure

Using the technique of traversing, the relative position of the control points is fixed by measuring the horizontal angle at each point, between the adjacent stations, and the horizontal distance between consecutive pairs of stations. Follow the following procedure for establishing control points using traversing.

7.4.1 Reconnaissance

Reconnaissance is a vital part of any survey project. Its purpose is to decide the best location for the traverse points. Successive points in the traverse must be inter-visible to make observations possible. If the purpose of the control network is the mapping of topographic detail only, then the survey points should be positioned to afford the best view of the terrain, thereby ensuring that the maximum amount of detail can be surveyed from each point.

7.4.2 Measurement Procedure for a Link Traverse

The link traverse start with two known points and ends with the other two known points and the measurement method is Face Left - Face Left - Face Right - Face Right (LLRR). The points that you are going to establish should be **permanent**, marked with paint with the name written next to them. **Note: The points should be inter-visible and located so that all the interesting data can be collected.** The major steps for link traverse are the following:

1. Level and center the instrument for example at B
2. In the main menu choose **"Known Back Sight Point"** method and insert the control points' coordinates
3. Measure at point A Face Left, then measure point 1 with Face Left
   Note: Record the azimuth and distance measurements manually for later calculation.
4. Change the Face to Right and measure point 1 then measure point A with the same Face.
   Note: You have to enter the same Point ID for both left and right face measurement of a certain point.
5. Then move the instrument to point 1, level and center it just as before and repeat the same procedure in order to measure the points 2 and 3.
6. Then finally set up the instrument at the known point C and close the traverse by measuring with both faces at the known point D after observing backsight at point 3.

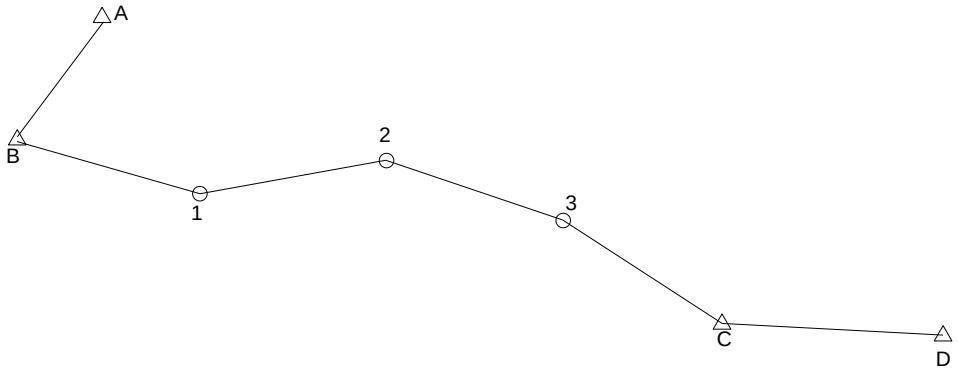


Figure 7-5. **Link traverse, where points A, B, C and D are known.**

7.4.3 Measurement Procedure for a Closed Traverse

It is required to plot a closed traverse ABCDE as shown in Figure 7-6.

7. Set up the theodolite instrument over station A and level it accurately instrument.
8. Use compass to get the North direction (use range pole)
9. Point to the North direction and set the horizontal angle to zero
10. Direct the telescope towards station B and observe the Azimuth of the line AB.
11. Measure distance AB using optical method or EDM.
12. Measure the included angle (or angle to the right) at A.
13. Shift the instrument from station A to station B and level it accurately and sight to A with the help of lower clamp screw. From the station B, measure distance BC and included angle at B.
14. Repeat 7 for the successive lines C, D, and E and measure distance of CD, DE and EA as well as included angle at C, D, and E.

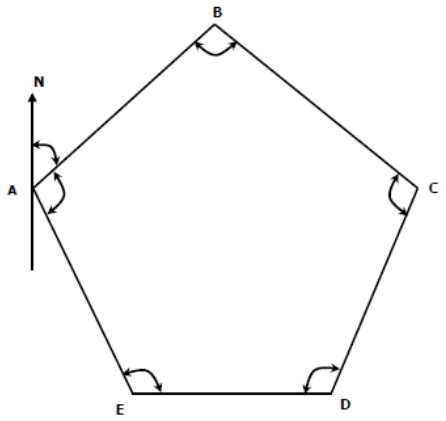


Figure 7-6 Example of closed traverse

7.5 Report Requirement

The report should include adjustment of the link and closed traverse and the calculated unknown coordinates in the standard reporting format.

Exercise

8

8 Intersection and Resection

8.1 Objective of the Experiment

The objective of this exercise is to familiarize yourself about establishment of horizontal control points using intersection and resection.

8.2 Theory

8.2.1 Intersection

This is a method of locating a control point without occupying it. On construction sites, prominent marks around a site, such as tall buildings, church spires and other clearly defined features may be useful as control points during construction. It is obviously not possible to set up an instrument at these but it is possible to obtain their coordinates by using intersection. Since they are usually in elevated positions they can be seen when the lines of sight to other control points at ground level become obscured as construction proceeds. In Figure 8-1, A and B are points in a control network with known coordinates (EA, NA) and (EB, NB). To coordinate unknown point P which lies at the intersection of the lines from A and B, a total station or theodolite is set up at A and B and the horizontal angles α and β are observed. The coordinates of P can be calculated by a number of different methods, two of which are given here.

In triangle ABP of Figure 8-1, the length and bearing of the baseline AB are obtained from their coordinates and are given by rectangular → polar conversions as

$$D_{AB} = \sqrt{\Delta E_{AB}^2 + \Delta N_{AB}^2} \qquad \theta_{AB} = \tan^{-1}\left[\frac{\Delta E_{AB}}{\Delta N_{AB}}\right]$$

where $\Delta E_{AB} = E_B - E_A$ and $\Delta N_{AB} = N_B - N_A$.

The sine rule gives

$$D_{BP} = \frac{\sin\alpha}{\sin(\alpha+\beta)} D_{AB} \qquad D_{AP} = \frac{\sin\beta}{\sin(\alpha+\beta)} D_{AB}$$

The whole-circle bearings in the triangle are given by

$$\theta_{AP} = \theta_{AB} + \alpha \qquad \theta_{BP} = \theta_{BA} - \beta$$

These distances and bearings are used to calculate the coordinates of P along AP using polar → rectangular conversions as

$$E_P = E_A + D_{AP} \sin\theta_{AP} \qquad N_P = N_A + D_{AP} \cos\theta_{AP}$$

The calculations are checked along BP using

$$E_P = E_B + D_{BP} \sin\theta_{BP} \qquad N_P = N_B + D_{BP} \cos\theta_{BP}$$

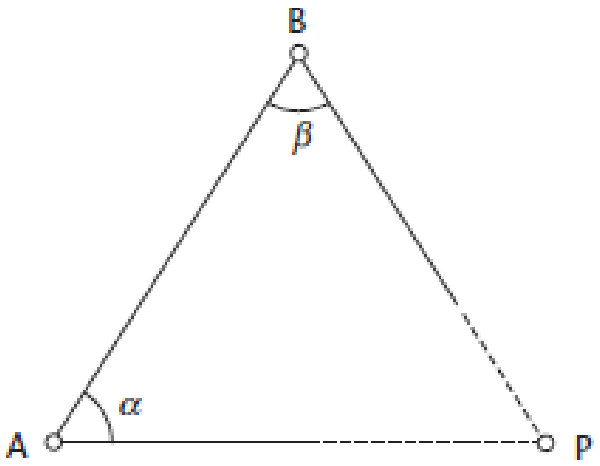


Figure 8-1 Intersection by angles

Intersection using the observed angles

If we adopt the *clockwise lettering sequence* used in Figure 8-1, the coordinates of P can be obtained directly from

$$E_P = \frac{(N_B - N_A) + E_A \cot\beta + E_B \cot\alpha}{\cot\alpha + \cot\beta}$$

$$N_P = \frac{(E_A - E_B) + N_A \cot\beta + N_B \cot\alpha}{\cot\alpha + \cot\beta}$$

8.2.2 Resection

This is a method of locating a point by taking observations from it to other known control points in a network. Two types of resection are possible – *angular resections* in which horizontal angles are

measured and *distance resections* in which horizontal distances are measured. Both of these are particularly useful for coordinating temporary control points on site which are called *free station points* because they can be established anywhere convenient.

Angular resections

These are used to coordinate a point by taking observations from it to existing control points – an advantage of the method is that a resection can be done without occupying any of the control points to which observations are taken. A good example of this occurs on high-rise buildings when ground control is no longer available or cannot be transferred through to the upper floors. In this case, intersected points can be fixed around the site prior to construction and used later as control for resections as the building is constructed.

A point can be coordinated in an angular resection by observing angles from it to at least three existing control points in a *three-point resection*, as shown in Figure 8-2. In this, horizontal angles α and β subtended at the resection point P by control points A, B and C are observed. If the triangles ABC of Figure 8-2 are *lettered clockwise*, the coordinates of P are given by

$$E_P = \frac{k_1 E_A + k_2 E_B + k_3 E_C}{k_1 + k_2 + k_3} \qquad N_P = \frac{k_1 N_A + k_2 N_B + k_3 N_C}{k_1 + k_2 + k_3}$$

where

$\frac{1}{k_1} = \cot \hat{A} - \cot \alpha$ α = clockwise angle between directions PB and PC

$\frac{1}{k_2} = \cot \hat{B} - \cot \beta$ β = clockwise angle between directions PC and PA

$\frac{1}{k_3} = \cot \hat{C} - \cot \gamma$ γ = clockwise angle between directions PA and PB

$\hat{A}$, $\hat{B}$ and $\hat{C}$ are the angles within the control triangle ABC.

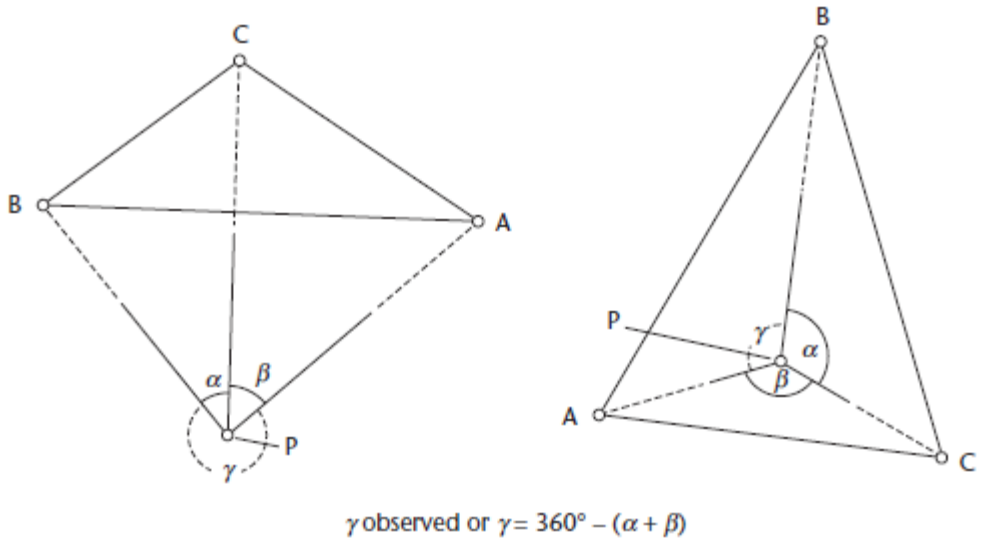


Figure 8-2 Resection figures

8.3 Instruments Used

Total station/Theodolite, Tripod, Tape, Reflector/Staff

8.4 Procedure

Intersection

To do this exercise, at least you need **two** control points (A, and B)

1. Set the Theodolite/Total Station at A and measure angle α which is the angle between the unknown point P and Point B as in Figure 8-1. Note when you measure direction to find α, read in both faces. (refer Exercise 3)

2. Repeat 1 by setting the Theodolite/Total Station at B to measure β.

3. Compute the coordinates of the unknown point P with appropriate formula

Resection

To do this exercise, at least you need **three** control points (A, B and C)

1. Set the Theodolite/Total Station at unknown point P

2. Target to known point B and C and measure angle α as in Figure 8-2 . Note when you measure direction to find α, read in both faces. (refer Exercise 3)

3. Repeat 2 by targeting point A and C to measure β
4. Repeat 2 by targeting point A and B to measure γ. (optional)
5. Compute the coordinates of the unknown point P with appropriate formula

8.5 Report Requirement

Prepare a standard report that include measurement procedure and computation for both methods.

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